ECE 310 Fall 2026
Lecture 3: Discrete-time systems

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Learning Objectives

- Define what represents a discrete-time system.
- State the conditions for systems to be linear, time-invariant, and causal.
- State the definition of a system being bounded-input bounded-output (BIBO) stable.
- Prove whether a given discrete-time system is linear, time-invariant, causal, or stable.



Discrete-time Systems

- A *discrete-time system* is any computational process that maps a discrete-time input, $x[n]$, to a discrete-time output, $y[n]$.
- Let T denote an abstract system or *operator* that maps $x[n]$ to $y[n]$.

$$x[n] \overset{T}{\mapsto} y[n]$$
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- Examples:

$$y[n] = x[n] - x[n-1]$$
$$y[n] = \text{median}\{x[n], x[n-1], x[n-2]\}$$
$$y[n] = \text{SnapchatFilter}(x[n])$$



Linearity

A discrete-time system T is *linear* if and only if it satisfies the following two conditions:

$$T(ax[n]) = aT(x[n]),\ a \in \mathbb{R} \qquad \text{(Homogeneity)}$$

$$T(x_1[n] + x_2[n]) = T(x_1[n]) + T(x_2[n]) \qquad \text{(Additivity)}$$



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In summary, we can prove a system is linear by showing it satisfies homogeneity **and** additivity; **or**, that the system satisfies superposition.



Linearity Practice

Ex. 1: $y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$

Ex. 2: $y[n] = x[n] + 1$



Extra Linearity Practice

Ex. 1: $y[n] = |x[n]|$

Ex. 3: $y[n] = e^{x[n]}$

Ex. 2: $y[n] = n(x[n] - x[n-1])$

Ex. 4: $y[n] = \max\{0, x[n]\}$



Time-invariance

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A discrete-time system T is *time-invariant* or *shift-invariant* if and only if

$$y[n - n_0] = T(x[n - n_0]),\ n \in \mathbb{Z}$$

for input $x[n]$ and output $y[n]$. In different notation, this may be written as

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In words, this means that shifting the input $x[n]$ by n_0 samples leads to a corresponding shift of n_0 samples in the output.



Time-invariance Practice

Ex. 1: $y[n] = x[n] - x[n-1]$

Ex. 2: $y[n] = x[2n]$



Extra Time-invariance Practice

Ex. 1: $y[n] = x[|n|]$

Ex. 3: $y[n] = e^{x[n]}$

Ex. 2: $y[n] = n(x[n] - x[n-1])$

Ex. 4: $y[n] = \max\{0, x[n]\}$



Causality

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A discrete-time system is **causal** if the system output only depends on **present** or **previous** input and output samples. In other words, the system does not depend on future samples.

Ex. 1: $y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$

Ex. 2: $y[n] = x[|n|]$



Extra Causality Practice

Ex. 1: $y[n] = x[|n| - n]$

Ex. 3: $y[n] = e^{x[n]}$

Ex. 2: $y[n] = (n+1)(x[n] - x[n-1])$

Ex. 4: $y[n] = \max\{0, x[n]\}$



BIBO Stability

BIBO stability

- A discrete-time system T is *bounded-input bounded-output stable* (BIBO stable) if and only if for any $x[n]$ such that $|x[n]| < \beta$ for all n, $|T(x[n])| < \alpha$ for all n where $0 \leq \alpha < \infty,\ 0 \leq \beta < \infty$.
- In words, a system is BIBO stable if and only if the system produces a bounded output signal for any bounded input signal. A signal is bounded if it takes a finite value for all indices $n \in \mathbb{Z}$.

Ex. 1: $y[n] = \sin^2(x[n])$

Ex. 2: $y[n] = \ln(|x[n]|)$



Extra Stability Practice

Ex. 1: $y[n] = \frac{x[n]}{|n|+1}$

Ex. 3: $y[n] = e^{x[n]}$

Ex. 2: $y[n] = n(x[n] - x[n-1])$

Ex. 4: $y[n] = \max\{0, x[n]\}$

